HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python**-based ETL pipelines to ingest and transform ~50GB of daily financial transaction data, reducing manual processing time by **40%**.
- Developed and maintained **Java Spring Boot** microservices, building **REST APIs** for inter-service communication and data access, handling an average of **1,500 requests per minute**.
- Engineered **SQL** queries and stored procedures for **PostgreSQL** and **Oracle** databases to support analytics and reporting, improving data retrieval efficiency for key operational reports by **18%**.
- Built and productionised **machine learning** models for anomaly detection using **scikit-learn** on a dataset of **~150,000** transaction records, achieving an **AUC of 0.82**.
- Developed and deployed **Generative AI** solutions incorporating **RAG-based architectures** and LLMs, integrating with modern cloud data platforms to provide intelligent application features.
- Containerised **Python** and **Java** applications using **Docker** and deployed them to a **Kubernetes** cluster (EKS), managing CI/CD pipelines with **Jenkins** for automated testing and deployment.
- Collaborated with data science and business teams to translate complex data challenges into scalable, secure, and commercially valuable **AI/ML** solutions, focusing on end-to-end ownership from design to deployment.

SKILLS

Languages – Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, scikit-learn, pandas, NumPy, FastAPI, LangChain

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, CI/CD, MLOps

Databases – Snowflake, PostgreSQL, MySQL, Oracle

AI/ML & GenAI – Machine Learning, Generative AI, LLMs, RAG, Model Deployment, Model Monitoring, Data Pipelines

PROJECTS

AI-Driven Anomaly Detection System

- Developed a **Python**-based system to detect anomalous financial transactions using **machine learning** techniques, processing approximately **100,000** records.
- Engineered feature extraction pipelines with **pandas** and **NumPy**, transforming raw transaction data into meaningful inputs for ML models.
- Trained and evaluated a **gradient-boosted classifier** model on a dataset of **~150,000** transaction records, achieving an **AUC of 0.82** and a precision of **0.75**.

Intelligent Document Processing with RAG

- Built a **RAG-based architecture** using **LangChain** to enable intelligent querying of a large document corpus, improving information retrieval accuracy by **30%**.
- Developed **Python** services to ingest, chunk, and embed documents into a **vector database**, preparing them for retrieval by a **LLM**.
- Deployed the solution as a **FastAPI** service, integrating with cloud data platforms and demonstrating its capability to answer complex queries based on the ingested documents.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)